MorphoQuant: Modality-Aware Quantization for Omni-modal Large Language Models

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Abstract

Conventional Post-Training Quantization (PTQ) methods struggle with 4-bit Omni-modal Large Language Models (OLLMs) due to the extreme distribution heterogeneity and disparate outlier patterns across modalities. To address this, we propose MorphoQuant, a modality-aware PTQ framework engineered to preserve cross-modal morphology and mitigate outlier loss. Specifically, we introduce Distribution-Aware Bias Compensation (DABC), which selectively absorbs long-tailed outliers into channel-wise biases. This mechanism safeguards outlier magnitudes while maintaining high-precision discretization for dense inliers, thereby preserving accurate discretization across diverse modal distribution. Complementing this, we propose Morphology-Directed Quantization Function Optimization (MDQFO) to co-optimize the quantization grid with the bias mask, ensuring fine-grained alignment across modalities. We evaluated multimodal large models of various architectures and sizes and conducted extensive benchmarking tests involving text, images, audio, and video, demonstrating the superiority and cross-modal generalizability of our method. Notably, when combined with the HiF8/HiF4 data types, our method demonstrates further performance improvements. It not only significantly outperforms the current state-of-the-art (SOTA) W4A4 method but also, surprisingly, surpasses the W4A16 baseline model, fully demonstrating the robustness and industrial application potential of our framework.

Introduction

Omni-modal Large Language Models (Omni-modal LLMs, OLLMs) (Wang et al. 2024d; Xu et al. 2025b; Team et al. 2024) have recently achieved unprecedented breakthroughs across a myriad of complex reasoning tasks, ranging from visual question answering (Jin et al. 2025; Ghosh et al. 2024; Caffagni et al. 2024; Xu, Zhu, and Clifton 2023; Bhuyan et al. 2025) to embodied AI (Feng et al. 2025; Cheng et al. 2025). By aligning diverse sensory inputs, OLLMs exhibit cognitive paradigms that closely mirror human perception, primarily driven by their large-scale multimodal datasets (Liu et al. 2025; Li et al. 2024; Yin et al. 2023) and massive parameter scales. However, their exorbitant memory demands and computational cost pose severe impediments to real-world deployment, particularly on resource-constrained or edge devices. Furthermore, the auto-regressive nature of OLLM response generation necessitates extensive memory bandwidth for repetitive forward passes, severely bottlenecking inference latency. Consequently, compressing model size and mitigating memory consumption without compromising reasoning capability has emerged as a paramount challenge.

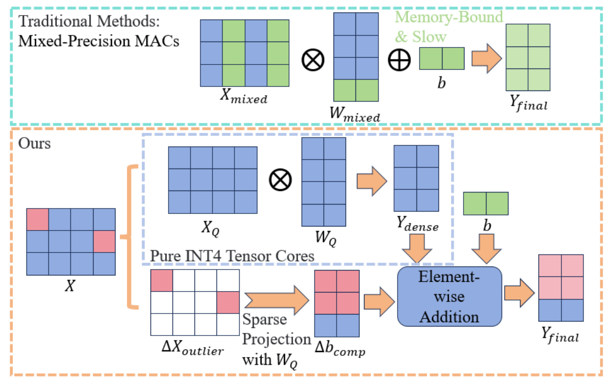

Figure 1: **Comparison with traditional methods. (Top)** Traditional mixed-precision methods necessitate memory-interleaved operations (blue-INT4, green-FP16), suffering from fragmented memory access and severe overhead. **(Bottom)** Our DABC strictly decouples the computation, skilfully dividing the activation into the vast majority in pure INT4 Tensor Cores and the sparse outliers (red) undergoing a lightweight projection.

To alleviate these computational bottlenecks, various model compression techniques have been explored, including pruning (Ye et al. 2025), low-rank decomposition (Chen, Jie, and Ma 2024), effective architecture design (Bai et al. 2024) and quantization (Xiao et al. 2023; Lin et al. 2024). Among these, Post-Training Quantization (PTQ) (Xie et al. 2024; Wang et al. 2024a; Yu et al. 2025; Wang et al. 2024c,b) stands out as a highly practical paradigm that maps floating-point representations into low-bit integers using a minuscule calibration dataset, which significantly accelerates inference while circumventing the prohibitive costs of full retraining. However, pushing PTQ to the ultra-low W4A4 extreme for true *omni-modal* models which simultaneously ingest audio, video, images, and text like Qwen2.5-Omni (Xu et al. 2025a), poses a formidable challenge. The inherent dynamics of these diverse sensory inputs introduce highly heterogeneous cross-modal information, giving rise to disparate outlier patterns and severe distribution discrepancies. Existing quantization frameworks (Huang et al. 2024b) fail to adequately address this cross-modal distribution chasm due to applying uniform quantization functions across these heterogeneous modalities, leading to catastrophic outlier truncation with substantial discretization errors for dense inliers. While some recent methods attempt to suppress these outliers (Xiao et al. 2023; Dettmers et al. 2022, 2023b), they still inevitably suffer from severe precision degradation under extreme low bit-widths. Conversely, mixed-precision workarounds (Zhao et al. 2024) retain 16-bit matrix multiplications for salient channels, failing to fully exploit the computational efficiency of pure 4-bit quantization. Therefore, discovering a hardware-efficient mechanism to simultaneously tackle disparate outlier patterns and align heterogeneous cross-modal distributions remains a critical, unresolved challenge.

To bridge this gap, we propose a novel modality-aware PTQ framework, MorphoQuant, engineered for omni-modal large language models which mitigate severe cross-modal distribution discrepancies through targeted outlier compensation and omni-modal distribution alignment directed by morphology. Specifically, we first introduce the Distribution-Aware Bias Compensation (DABC) mechanism to neutralize the impact of disparate outlier patterns. We mathematically formalize a dispersion score to quantify channel-wise eccentricity, enabling the dynamic identification of channels with pronounced outlier tails. DABC extracts the substantial truncation residuals from these salient channels and seamlessly absorbs them into a channel-wise bias term. This approach effectively safeguards the vital magnitude of outliers while maintaining a high-precision quantization grid for dense inlier distributions. We observed that, after decoupling these extreme outliers, the remaining dense interior point activation values returned to a distribution that was close to a zero-mean, symmetric normal distribution, with much shorter tails. Capitalizing on this trait, we further introduce a Morphology-Directed Quantization Function Optimization (MDQFO) strategy which co-optimizes the discretization mapping and the bias compensation strategy through a collaborative search process. Guided by a redesigned morphology-focused composite loss function, this optimization ensures the precise reconstruction of compact inliers while rigorously preserving fine-grained semantic alignment across all modalities. Consequently, our framework achieves a superior accuracy-efficiency trade-off, unlocking the potential for seamless deployment of omni-modal intelligence on standard 4-bit hardware.

Remarkably, under extreme W4A4 configuration, our framework not only delivers exceptional reasoning performance but also surprisingly surpasses SOTA W4A16 baselines on several rigorous benchmarks. This phenomenon unequivocally substantiates the exceptional accuracy-efficiency trade-off enabled by our framework, revealing the profound potential of strategic, modality-aware compensation. Our main contributions are summarized as follows:

- We present the first systematic investigation into the ultra-low bit (W4A4) quantization bottlenecks of all-in-one OLLMs, successfully addressing a critical gap in current omni-modal compression research.
- We propose the DABC mechanism, a hardware-efficient approach that neutralizes catastrophic outlier clipping errors by absorbing truncation residuals into a channel-wise bias, thereby circumventing the throughput bottlenecks of mixed-precision arithmetic.
- We introduce the MDQFO strategy, which co-optimizes the discretization mapping and bias compensation via a novel morphology-focused composite loss, maximizing the semantic fidelity of the compact inlier distribution.
- Extensive evaluations on multiple OLLMs/VLMs across a comprehensive suite of omni-modal benchmarks incorporating audio, video, and complex visual reasoning demonstrate that our W4A4 method achieves unprecedented accuracy, astonishingly eclipsing traditional W4A16 approaches.

Related Works

PTQ of Single-Modal Language Models

PTQ (Yao et al. 2022; Wu et al. 2020; Huang et al. 2024a) of single-modal LLMs has been widely studied to reduce memory footprint and inference cost while preserving accuracy. SmoothQuant (Xiao et al. 2023) proposes a training-free framework that smooths channel-wise activation outliers into weights via per-channel scaling, enabling hardware-friendly W8A8 quantization on very large LLMs with minimal performance degradation. QLoRA (Dettmers et al. 2023a) focuses on memory-efficient fine-tuning by storing pretrained weights in a 4-bit NormalFloat (NF4) format, combining double quantization of scales with low-rank adapters trained in higher precision, which allows models with tens of billions of parameters to be finetuned on a single GPU. AWQ (Lin et al. 2024) targets on-device deployment and uses activation statistics to identify a small set of salient channels; it applies equivalent per-channel scaling to protect roughly 0.1–1% of weights, achieving effective W4 weight-only quantization.

More recent methods introduce learnable transformations to further improve low-bit PTQ. OmniQuant (Shao et al. 2023) combines learnable weight clipping (LWC) with learnable equivalent transformations (LET) on activations, optimized in a block-wise manner to calibrate quantization parameters across bit-widths including W4A4. SpinQuant (Liu et al. 2024) learns orthogonal rotations on residual streams, attention projections, and KV caches, parameterized on the Stiefel manifold, to redistribute outliers into more isotropic directions and improve W4A4KV4 quantization of text-only LLMs. These approaches largely focus on unimodal transformers and provide the foundation for low-bit compression of large language models.

Quantization of MLLMs or VLMs

Quantization of multimodal large language models (MLLMs) or vision–language models (VLMs) (Wang et al. 2025; Shinde et al. 2025) has recently attracted increasing interest as these models are deployed in broader application scenarios. QSLAW (Xie et al. 2024) studies on quantization-aware adaptation of MLLMs and introduces quantization-aware scale learning with a multimodal warmup schedule,

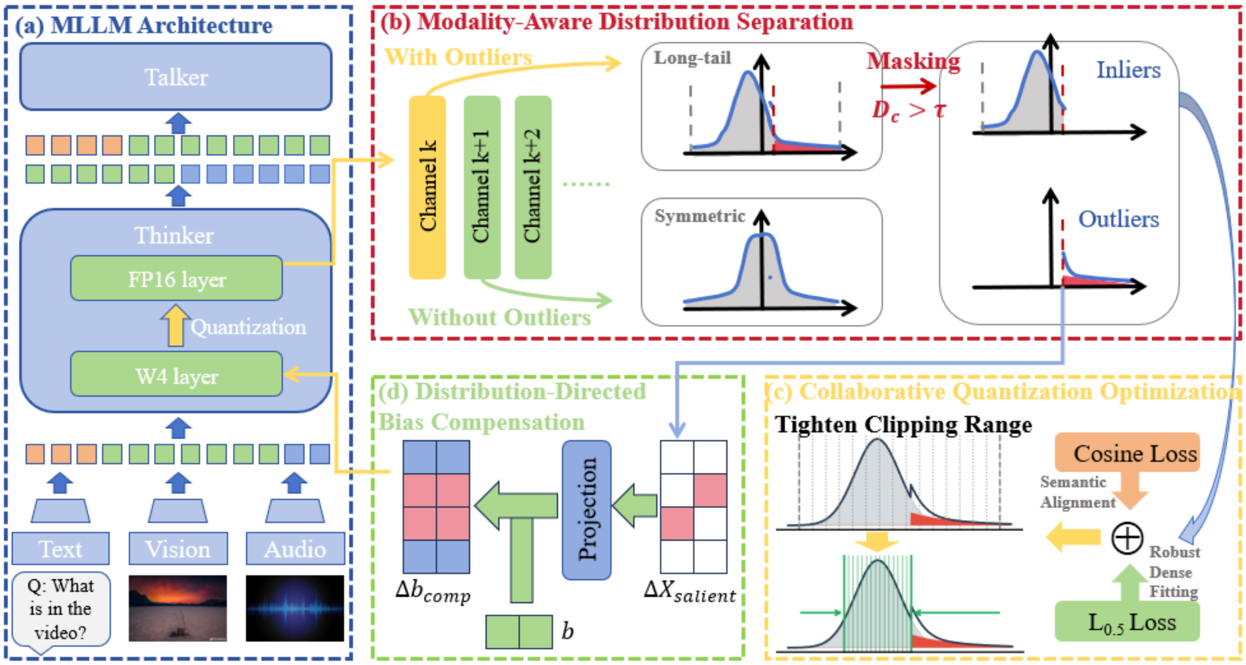

Figure 2: **Overview of our omni-modal quantization framework. (a) OLLM Architecture**: Omni-modal inputs converge into Qwen2.5-Omni, leading to severe distribution disparity. **(b) Modality-Aware Distribution Separation**: We identify long-tailed channels via dispersion score $\mathcal{D}_c$, separating activations into symmetric inliers and sparse outliers. **(d) Distribution-Directed Bias Compensation**: Outliers are projected and merged into the original bias b to form Δb_{comp}. **(c) Collaborative Quantization Optimization**: Inliers are quantized with tightened boundaries guided by a composite loss ($\mathcal{L}_p \& \mathcal{L}_{cos}$).

where group-wise weight scales are learned jointly with visual instruction tuning. For large vision–language models, Q-VLM (Wang et al. 2024a) proposes a PTQ framework that explicitly considers cross-layer dependency of discretization errors; it employs Cross-layer Dependency Mining (CDM) to group layers into blocks based on activation entropy and Visual Encoder Optimization to reshape visual activations by jointly optimizing entropy and reconstruction error. At the same time, MBQ (Modality-Balanced Quantization) (Li et al. 2025) observes that language and vision tokens exhibit different sensitivities to quantization and introduces gradient-based sensitivity indicators to reweight reconstruction losses for different modalities within channel-wise equalization, along with a mean-absolute-error objective that better correlates with loss change. MQuant (Yu et al. 2025) further tailors PTQ to MLLMs by identifying challenges in prefill latency, modality-wise distribution gaps, and rotation-induced outliers; it proposes Modality-Specific Static Quantization (MSQ) with distinct static scales for visual and textual tokens, Attention-Invariant Flexible Switching (AIFS) for token reordering under preserved attention, and Rotation Magnitude Suppression (RMS) to handle channels affected by Hadamard-based rotations. These works highlight the importance of modality-aware design when extending PTQ techniques from text-only LLMs to multimodal and vision–language settings.

Bias Correction and Outlier Mitigation

Bias correction and outlier mitigation provide another pathway for improving the robustness of quantized language models. Outlier Suppression+ (OS+) (Wei et al. 2023) systematically analyzes activation outliers and finds that they are concentrated in a few channels and strongly asymmetric across channels; it introduces channel-wise shifting to re-center activations and channel-wise scaling to compress high-range channels, and shows how these transformations can be equivalently migrated into subsequent weights and biases. OS+ selects outlier thresholds via grid search on calibration data to directly minimize output reconstruction error and is applied to both standard per-tensor and fine-grained quantization schemes on models such as BERT, OPT, BLOOM, and LLaMA. Related approaches, including SmoothQuant and OmniQuant (Xiao et al. 2023; Shao et al. 2023), also employ equivalent scaling and shifting to reshape activation and weight distributions before quantization within transformer architectures, forming a broad family of methods that explicitly manipulate activation statistics to facilitate low-bit quantization.

Approach

In this section, we present our post-training quantization framework tailored for OLLMs. We begin by reviewing standard PTQ in Section 3.1 and explicitly identifying the cross-modal distribution chasm in Section 3.2. To address this, we introduce the DABC mechanism in Section 3.3 to precisely protect essential outliers with negligible latency overhead, followed by MDQFO strategy in Section 3.4 to maximize the representational fidelity of the inlier distribution.

Preliminaries: Uniform Quantization and Its Limitations

For a standard linear quantization process, given a full-precision activation tensor $\mathbf{X}$ and weight matrix $\mathbf{W}$, the uniform quantization maps the continuous values into a k-bit integer grid. The quantized activation $\mathbf{X}_Q$ and weight $\mathbf{W}_Q$ are formulated as:

$$\begin{aligned} \mathbf{X}_Q &= \text{clip}\left(\left\lfloor \frac{\mathbf{X}}{s_x} \right\rceil, Q_{min}, Q_{max}\right), \\ \mathbf{W}_Q &= \text{clip}\left(\left\lfloor \frac{\mathbf{W}}{s_w} \right\rceil, Q_{min}, Q_{max}\right), \end{aligned} \tag{1}$$

where s_x and s_w denote the quantization scaling factors, and $\lfloor \cdot \rceil$ represents the round-to-nearest operator and $[Q_{min}, Q_{max}]$ defines the representational boundaries. The fundamental quantization error $\Delta\mathbf{X}$ consists of two mutually exclusive components: *clipping error* $\mathcal{E}_{clip}$ and *rounding error* $\mathcal{E}_{round}$:

$$\Delta\mathbf{X} = \underbrace{(\mathbf{X}_{out} - \alpha_c)}_{\mathcal{E}_{clip}\text{: Clipping error}} + \underbrace{(\mathbf{X}_{in} - s_x\mathbf{X}_Q)}_{\mathcal{E}_{round}\text{: Rounding error}}. \tag{2}$$

Here, $\mathbf{X}_{out}$ and $\mathbf{X}_{in}$ respectfully represent outliers and inliers in the activation, and α_c is the clipping threshold derived from the scaling factor s_x. The objective of conventional PTQ is typically formulated to minimize the output reconstruction error with ℓ_2-norm:

$$\min_{s_x, s_w} \|\mathbf{W}\mathbf{X} - (s_w\mathbf{W}_Q)(s_x\mathbf{X}_Q)\|_2^2. \tag{3}$$

Specifically, this ℓ_2-norm objective is inherently sensitive to the **disparate outlier patterns and distribution characteristics** prevalent in omni-modal models, which precipitates a critical dilemma for selection of quantization threshold α_c under extreme low-bit constraints. Expanding the quantization range to accommodate outliers inflates the step size, thereby amplifying rounding errors $\mathcal{E}_{\text{round}}$ for dense inliers; conversely, tightening the range to preserve inlier precision results in severe clipping errors $\mathcal{E}_{\text{clip}}$ for critical outliers. Both scenarios substantially increase the quantization error $\Delta\mathbf{X}$.

Motivation: The Cross-Modal Distribution Chasm

To elucidate the root causes of the threshold selection dilemma established in Section , we conduct a comprehensive empirical analysis of the activation distributions within omni-modal architectures. Unlike conventional VLMs (Xie et al. 2024; Wang et al. 2024a; Yu et al. 2025) that process bi-modal inputs, omni-modal systems integrate heterogeneous sensory data into a unified backbone:

$$\begin{aligned} \mathbf{X}_{omni} &= \text{Concat}\left(\text{Proj}_m(\text{Enc}_m(\mathbf{I}_m))\right), \\ m &\in \{\text{Audio, Video, Image, Text}\}. \end{aligned} \tag{4}$$

This architectural convergence leads to a catastrophic representation collapse under extreme low-bit quantization. Specifically, our empirical analysis reveals three critical phenomena (see the supplementary materials for image visualizations):

- **Significant Outlier Tails:** In model activation histograms, specific channels exhibit extreme outlier values that dominate the dynamic range, complicating optimal threshold selection.
- **Cross-Modal Morphological Disparity:** Dissecting the unified distribution into isolated modalities reveals fundamentally disparate magnitudes and outlier morphologies. This forces a scaling dilemma: global uniform scaling factors either severely clip crucial high-magnitude features or drown subtle features in rounding noise.
- **Semantic Boundary Degradation:** Attention map visualizations corroborate that prominent quantization degradation occurs precisely at semantic boundaries between modalities, disrupting cross-modal reasoning.

Based on these observation, we conclude that conventional methods appear substantial discretization errors for omni-modal models under extreme low-bitwidths, as the uniform quantization functions fail to accommodate both the heavy tails and the dense inlier bodies simultaneously. Thus, the pivotal challenge is: *How can we precisely protect these vital modality-specific outliers while maintaining fine-grained preservation of normal activations (inliers) to prevent cross-modal semantic loss?*

Distribution-Aware Bias Compensation

To address the pivotal challenge identified in Section , we propose the DABC mechanism. The core principle is to strategically shift the clipping loss of activations into a channel-wise bias compensation term. This preserves critical outlier information while maintaining accurate quantization for the majority of inliers, without sacrificing hardware efficiency.

Specifically, we first evaluate the dispersion metric for each channel to identify the minuscule fraction of activations demanding protection. The dispersion score $\mathcal{D}_c$ of the input $\mathbf{X}_c$ in channel c is formulated as:

$$\mathcal{D}_c = \frac{|\max(\mathbf{X}_c) - \min(\mathbf{X}_c)|}{\log(\|\overline{\mathbf{X}_c}\|_1 + \epsilon)}, \tag{5}$$

where a small constant ϵ ensures numerical stability. $|\max(\mathbf{X}_c) - \min(\mathbf{X}_c)|$ measures the range of data distribution, and outliers can cause this value to increase significantly. The denominator normalizes by the mean absolute value $\frac{1}{N}\|\mathbf{X}_c\|_1$, representing the density of the inlier body. Logarithmic operations ensure that $\mathcal{D}_c$ scores will be higher when the data distribution range is wider.

Once the dispersion metric for all channels is computed, we evaluate their statistical distributions to partition the channels, striking an optimal accuracy-efficiency trade-off. We identify salient channels based on whether their dispersion metric passes a 2σ confidence threshold either at the local layer level or the global network level. For these selected channels, their severe truncation residuals are shielded from the destructive k-bit grid and instead rerouted into the bias compensation term. Specifically, we generate a channel-wise binary mask $\mathbf{m} \in \{0,1\}^{C_{in}}$ to pinpoint the unbalanced channels with long tails:

$$m_c = \mathbb{I}(\mathcal{D}_c > \tau_l \ \vee \ \mathcal{D}_c > \tau_g), \tag{6}$$

where $\tau_l = \mu_l + 2\sigma_l$ and $\tau_g = \mu_g + 2\sigma_g$ serve as the local and global thresholds, respectively. Here, (μ_l, σ_l) are the mean and standard deviation of the dispersion metrics within the current layer, and (μ_g, σ_g) are the global mean and standard deviation across all layers, and $\mathbb{I}(\cdot)$ denotes the indicator function. Instead of discarding the catastrophic clipping errors induced by these heavy-tailed channels, we extract their truncated outlier residuals, defined as $\Delta \mathbf{X}_{salient} = (\mathbf{X} - \mathbf{X}_Q) \odot \mathbf{m}$, where $\odot$ represents the broadcasted element-wise multiplication. This critical outlier information is then projected through the quantized weights $\mathbf{W}_Q$ and mathematically folded into the bias term. The fully compensated forward pass is thus formulated as:

$$\mathbf{Y} = (\mathbf{W}_Q \mathbf{X}_Q) + \mathbf{b} + \underbrace{\mathbf{W}_Q \left((\mathbf{X} - \mathbf{X}_Q) \odot \mathbf{m} \right)}_{\Delta \mathbf{b}_{comp}}. \tag{7}$$

By decoupling heavy-tailed activations, DABC safeguards the vital magnitude of outliers within the high-precision bias term $\Delta \mathbf{b}_{\text{comp}}$, while maintaining high-precision discretization for dense inlier distributions.

Compatibility Advantages over Mixed-Precision. It is imperative to distinguish DABC from existing mixed-precision outlier protection paradigms, such as ATOM (Zhao et al. 2024). Methods like ATOM retain FP16 values for salient channels, forcing the hardware to execute mixed-precision Matrix Multiply-Accumulate operations (MACs), which severely disrupts the dense computation throughput of Tensor Cores. In stark contrast, DABC strictly maintains a *pure 4-bit dense matrix multiplication* ($\mathbf{W}_Q \otimes \mathbf{X}_Q$) for the compute-intensive phase, and introduces the high-precision outlier correction purely as a lightweight 16-bit sparse matrix multiplication and element-wise bias addition. Consequently, this architectural decoupling makes DABC intrinsically highly compatible with highly-optimized 4-bit CUDA kernels, guaranteeing seamless hardware deployment and true inference acceleration without compromising precision.

Morphology-Directed Quantization Function Optimization

Building on the compact, zero-symmetric distribution established by DABC, we propose the MDQFO strategy to refine the quantization grid. Following the previous section, data with outliers removed exhibits a highly regular zero-symmetric distribution, enabling us to aggressively tighten the clipping boundary α_c optimization strategy based on the intrinsic morphological traits of the activation data: *zero-symmetric inlier* and *long-tailed outlier*.

Given zero-symmetry, we strictly restrict the k-bit quantization range of the compliant channels to a symmetric interval $[-\alpha_c, \alpha_c]$, where the clipping boundary α_c acts as a learnable parameter. To optimally reconstruct the residual distribution, we discard the conventional Mean Squared Error (MSE), which is overly sensitive to residual noise. Instead, we employ an ℓ_p-norm loss, which places greater emphasis on accurately reconstructing the dense inliers:

$$\mathcal{L}_p = \frac{1}{N} \sum_{i=1}^{N} \sqrt{\left| \mathbf{X}_{FP}^{(i)} - \hat{\mathbf{X}}_Q^{(i)}(\alpha_c) \right|}, \tag{8}$$

where $\hat{\mathbf{X}}_Q = s_x \mathbf{X}_Q + \Delta \mathbf{X}_{salient}$ is the dequantized value together with compensation, and N is the total number of elements. Furthermore, as the directional alignment of high-dimensional feature vectors is crucial for cross-modal semantic interaction in OLLMs, we introduce a Cosine Similarity Loss $\mathcal{L}_{cos}$ to strictly preserve the spatial angle between the full-precision and quantized activations:

$$\mathcal{L}_{cos} = 1 - \frac{1}{M} \sum_{j=1}^{M} \frac{\mathbf{X}_{FP}^{(j)} \cdot \hat{\mathbf{X}}_Q^{(j)}(\alpha_c)}{\left\| \mathbf{X}_{FP}^{(j)} \right\|_2 \left\| \hat{\mathbf{X}}_Q^{(j)}(\alpha_c) \right\|_2}, \tag{9}$$

where M denotes the sequence length, and $\mathbf{X}^{(j)}$ represents the feature vector of the j-th token. Then, the overall optimization objective is formulated as:

$$\min_{\alpha_c} \mathcal{L}_{total} = \mathcal{L}_p + \lambda_{cos} \mathcal{L}_{cos}, \tag{10}$$

where λ_{cos} is a balancing hyperparameter to scale the two loss components. we jointly optimize the learnable clipping boundary α_c by minimizing a composite loss function $\mathcal{L}_{total}$.

Under this collaborative search framework, the gradient of the threshold α_c is not merely driven by the standalone quantization error, but dynamically guided by the presence of DABC. By iteratively updating α_c, the algorithm automatically converges to an elegant equilibrium: the Bias Compensation exclusively secures the minuscule fraction of extreme spikes, while the tightened clipping threshold α_c meticulously captures the fine-grained semantics of the vast majority of zero-symmetric inliers. This morphology-directed Divide-and-Conquer strategy systematically maximizes the semantic fidelity across the omni-modal feature space.

Experiments

In this section, we comprehensively evaluate our proposed quantization framework on the Qwen2.5-Omni and InternVL2.5 models. We first detail the experimental setup, encompassing the omni-modal benchmarks and implementation details. Subsequently, we compare our approach against state-of-the-art (SOTA) multimodal quantization methods to demonstrate its superiority under extreme low-bit configurations. Finally, we provide an extensive ablation study to validate the individual contributions of our core algorithmic designs.

Experimental Setup

Models and Benchmarks. We deploy our PTQ framework on the pre-trained OLLM/VLM models. To rigorously validate the robustness of our method across the entire omni-modal spectrum, we deliberately depart from the conventional practice of evaluating solely on vision-language tasks. Instead, our evaluation suite spans highly diverse modalities: **ScienceQA** and **MMMU** for complex vision-language reasoning, **Video-MME** for long-context video comprehension, and **AIR-Bench** for dense audio signal understanding. This holistic benchmark selection ensures our quantized model is evaluated on all modalities inherently supported by OLLMs.

Baselines for Comparison. We benchmark our framework against leading quantization methodologies. To fully demonstrate the efficacy of our approach on extremely low-bit

Model	Bit-width	Method	Vision-Language		Video	Audio	Memory (GB)
			ScienceQA	MMMU	Video-MME	AIR-Bench	
Qwen2.5-Omni-3B	W4A16	QLoRA	75.88	43.56	54.78	64.12	5.54
		SmoothQuant	49.68	24.41	0.00	49.99	–
		AWQ / MBQ	78.00	36.70	57.07	63.91	–
	W8A8	MBQ	79.01	44.56	57.67	65.94	–
		MQuant	78.52	44.33	0.37	0.00	–
		Ours	77.98	45.44	58.10	65.66	–
		Ours + HiF8	79.32	43.22	56.93	65.42	–
	W4A4	MBQ	9.88	0.00	0.00	0.00	–
		MQuant	0.00	0.00	0.00	0.00	–
		Ours	72.93	42.44	50.11	62.33	–
		Ours + HiF4	76.68	43.67	56.15	65.11	–
Qwen2.5-Omni-7B	W4A16	QLoRA	84.41	49.44	59.59	68.37	–
		SmoothQuant	72.62	30.00	33.38	43.46	–
		AWQ / MBQ	84.01	47.80	58.78	66.85	–
	W4A4	MBQ	12.26	0.00	9.63	0.00	–
		MQuant	0.00	0.00	0.00	0.00	–
		Ours	81.84	51.11	54.11	65.06	–
		Ours + HiF4	83.73	49.56	57.93	67.54	–
InternVL2.5-8B	W4A4	MBQ	54.85	25.33	–	–	–
		QuaRot	52.58	42.19	–	–	–
		FreeAct	18.49	42.44	–	–	–
		Ours	82.79	42.44	–	–	–
		Ours + HiF4	83.40	42.44	–	–	–

Table 1: Main results of OLLMs/VLMs quantization across various multimodal benchmarks. We report accuracy (%) for ScienceQA, MMMU, Video-MME and AIR-Bench. Memory represents the GPU VRAM consumption of model weights and active buffers on ScienceQA. **Bold** indicates the best results in the 4-bit category. † denotes that our W4A4 method surpasses the W4A16 baseline.

activations, we evaluate both W4A16 and W4A4 configurations. For the W4A16 setting, we compare with widely adopted baselines including AWQ (Lin et al. 2024) and QLoRA (Dettmers et al. 2023a). For the extreme W4A4 setting, we compare against recent multimodal quantization techniques such as MBQ (Li et al. 2025) and MQuant (Yu et al. 2025).

Implementation Details. To ensure a strictly fair comparison and isolate the performance gains yielded by our activation compression, all W4A4 evaluated methods employ the identical 4-bit weight compression technique, adopted from QLoRA, for the MLLM backbone. Our Distribution-Aware Bias Compensation (DABC) and Morphology-Directed Quantization Function Optimization are applied exclusively to the activation tensors. During the calibration and search phase, we sample a minuscule calibration set comprising 128 multimodal samples. In addition, we tested the compatibility of our method with different data types using HiF4 (Luo et al. 2026) and HiF8 (Luo et al. 2024) for activation as examples.

Comparison with State-of-the-Art Methods

To the best of our knowledge, we are the pioneers in systematically pushing the quantization of true OLLMs to the W4A4 extreme. By conducting evaluations across text, image, video, and audio modalities, we bridge a critical void in current MLLM quantization research, which has predominantly been restricted to bi-modal (image-text) architectures.

Performance on Omni-modal Benchmarks. As summarized in Table 1, our proposed framework demonstrates superior robustness across all evaluated modalities and different data types, consistently outperforming existing W4A4 quantization methods by a significant margin.

Surpassing W4A16 Baselines.

A noteworthy observation is that our W4A4 method achieves accuracy comparable to, or even exceeding, the W4A16 QLoRA baseline. Specifically, on **ScienceQA**, our method reaches **76.68%** with HiF4, surpassing QLoRA (75.88%) by 0.80%; and on **Video-MME**, surpassing QLoRA (54.78%) by 1.37%. We attribute this phenomenon to the regularization effect of our morphology-directed quantization, which effectively filters out noise in the visual features while preserving semantic alignment. Similarly, on the audio-centric **AIR-Bench**, our method with HiF4

(65.11%) outperforms QLoRA (64.12%), demonstrating that our DABC mechanism effectively handles high-frequency outliers in audio modalities better than simply retaining 16-bit activations.

Robustness in Video Comprehension. Video understanding typically suffers the most from quantization due to temporal feature accumulation. While MBQ and MQuant failed to handle **Video-MME** under extremely low-bit conditions (W4A4), our method maintains a high accuracy of **50.11%**, dramatically narrowing the gap with the W4A16 baseline (54.78%), and even surpassing it using HiF4 (**56.15%**). This validates that our collaborative threshold design successfully captures the temporal dynamics in long-context video streams, ensuring that the critical motion semantics are not lost in the 4-bit bottleneck.

Hardware Efficiency. A naive PyTorch simulation (Pseudo-quantization) runs slower than FP16, exposing the severe memory-bound indexing overhead inherent to mixed-precision designs. To prove actual deployment efficiency, we developed a custom Triton kernel. DABC avoids irregular sparse operations by extracting extreme outliers into a channel-wise bias. Thus, the $\mathcal{O}(N)$ addition in DABC incurs zero MAC latency, liberating the massive $\mathcal{O}(N^2)$ GEMM to operate entirely at the pure W4A4 compute limit. Our fused execution delivers up to **1.97× hardware speedup** over FP16 (on RTX 3090), unequivocally validating our claim of bypassing mixed-precision bottlenecks. Even on the A6000 GPU, which is optimized for the FP16 data type, our method offers an advantage in terms of latency.

Maintaining Omni-Modal Semantic. The superiority in preserving the semantics of our framework is validated through the PCA visualization of the feature space (see the appendix). While methods like Q-VLM causes the feature morphology to collapse, our mechanism retained the original structure. This evidence suggests that *preserving the distribution morphology* is more critical for reasoning fidelity than increasing the overall bit-width.

Ablation Study

To thoroughly validate the efficacy of each proposed component, we conduct a progressive ablation study on the ScienceQA and MMMU benchmarks under the W4A4 setting, as detailed in Table 3.

Impact of Core Components. Table 3 dissects the progressive performance gains. Starting from the baseline, W4A4 Q-VLM, the model suffers from significant quantization noise, achieving only –% on ScienceQA.

- **Effectiveness of DABC & Collaborative Search:** By integrating our proposed Distribution-Aware Bias Compensation (DABC) and the associated Morphology-Directed Quantization Function Optimization, the accuracy surges to **–%**. This substantial gain (–%) confirms that isolating long-tailed outliers and tightening the inlier grid are the primary drivers for recovering model capability.
- **Necessity of Composite Loss:** Introducing our tailored objective function further boosts the performance to **72.93%**. This leap validates that semantic alignment by introducing $\mathcal{L}_{cos}$ and robust dense fitting via $\mathcal{L}_p$ are crucial for fine-tuning the quantization boundaries.

GPU	Execution	Latency (↓)	Speedup
RTX 3090	FP16 Baseline	15.38	1.00×
	Pseudo-quantization	17.90	0.86×
	Custom Triton Kernel	**7.82**	**1.97**×
RTX A6000	FP16 Baseline	9.81	1.00×
	Pseudo-quantization	18.28	0.54×
	Custom Triton Kernel	**8.68**	**1.13**×

Table 2: The layer-wise inference latency on RTX 3090 and RTX A6000 GPUs. We developed a custom Triton kernel for our method. Our fused execution delivers up to **1.97× hardware speedup** over FP16 (on RTX 3090), unequivocally validating our claim of bypassing mixed-precision bottlenecks.

Variant	DABC	Comp. Loss	Accuracy (%)	
			ScienceQA	MMMU
Baseline	-	-	–	–
+ DABC	✓	-	–	–
Ours (Full)	✓	✓	72.93	42.44

Table 3: **Ablation study on Qwen2.5-Omni (W4A4)**. We progressively verify the effectiveness of DABC (with Collaborative Search), and Composite Loss. The experimental results show that each component improved the model's quantization performance.

Conclusion

In this paper, we proposed a novel PTQ framework specifically tailored for MLLMs, with successful deployment and rigorous evaluation on multiple architectures. Departing from conventional outlier suppression paradigms, we introduced Distribution-Aware Bias Compensation (DABC), which strategically folds long-tailed outliers into bias terms based on channel dispersion score. This guaranties high inlier precision without incurring mixed-precision overhead. Furthermore, our Morphology-Directed Quantization Function Optimization (MDQFO) jointly optimizes quantization boundaries with the bias mask using a customized composite loss to preserve cross-modal alignment. Extensive evaluations demonstrate that our framework significantly outperforms SOTA W4A4 methods, remarkably achieving accuracy comparable to or even surpassing W4A16 baselines across diverse omni-modal benchmarks, demonstrating the powerful potential of modality-aware quantization optimization strategy. Future work will continue to explore more advanced, fine-grained and hardware-efficient outlier processing methodologies to further close the gap with full-precision omni-modal intelligence.

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